

Agent-based Simulation with Mobility Personas

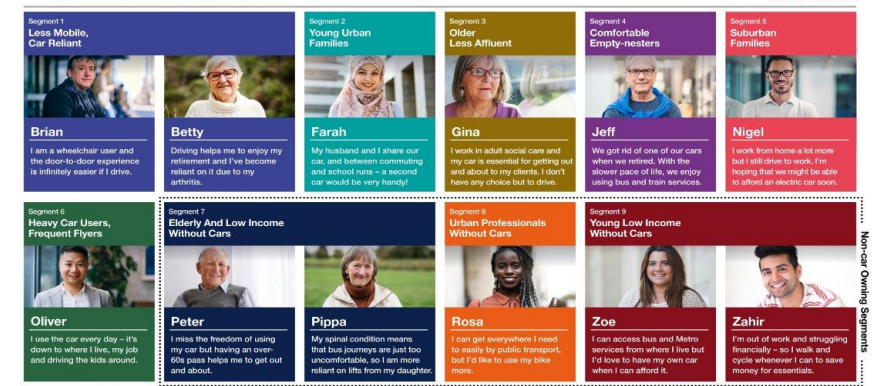
An LLM-based Persona Labelling Approach for MATSim Synthetic Populations

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MATSim User Meeting 2026

Meet the Personas



© Non-car owners segments (7-9). Images used in this document are of fictional people and have been chosen to portray the key characteristics of Personas.

v0 June 2020

28 Sep 2026

UK National Research Hub Transit



- Duration: 2024 – 2029;
- 22 PDRAs and 16 PhDs
- 8 leading universities in the UK;
- 67 partners from research institutions, public and private sectors;
- Led by Heriot-Watt University and the University of Glasgow



Digital twin federations for UK transport decarbonation and integration of transport solutions



Running Agent-based Simulation

Agents compete and interact during MATSim run



Each agent has a profile



Each agent performs activities



Each agent responds to the network

Disaggregated individual decisions —→ Aggregated system performance

14.2.4 Travel

Travel disutility for a leg q is given as

$$S_{trav,q} = C_{mode(q)} + \beta_{trav,mode(q)} \cdot t_{trav,q} + \beta_m \cdot \Delta m_q + (\beta_{dist,mode(q)} + \beta_m \cdot \gamma_{dist,mode(q)}) \cdot d_{trav,q} + \beta_{transfer} \cdot x_{transfer,q} \quad (14.5)$$

Factors considered

Time, distance, fare...

Different preferences, similar choices(?)



Agent A

Young urban families with children



Agent B

People with mobility impairment tend to use cars as their primary mode of transport

Same observable travel context

Car available

PT journey: 40 min / £2

Car journey: 20 min / £15

Would they make the same choices?

Could segmentation help?

LITERATURE TELLS US

Attitudes and constraints shape travel choices

Segmentation provides a practical way to represent travel behavioural heterogeneity without understanding every individual independently

A standard segmentation process

Segment	Age	Gender	Income	Car	Household Size	Urban/Rural	Travel Frequency	Travel Mode	Travel Purpose	Travel Constraints	Travel Attitudes	Travel Behaviours
1	18-24	Male	Low	Yes	1-2	Urban	High	Public Transport	Work	Time, Cost	Convenience	Regular, Short Trips
2	25-34	Female	Low	No	1-2	Urban	High	Public Transport	Work	Time, Cost	Convenience	Regular, Short Trips
3	35-44	Male	Low	Yes	2-3	Urban	High	Public Transport	Work	Time, Cost	Convenience	Regular, Short Trips
4	45-54	Female	Low	No	1-2	Urban	High	Public Transport	Work	Time, Cost	Convenience	Regular, Short Trips
5	55-64	Male	Low	Yes	2-3	Urban	High	Public Transport	Work	Time, Cost	Convenience	Regular, Short Trips
6	65+	Female	Low	No	1-2	Urban	High	Public Transport	Work	Time, Cost	Convenience	Regular, Short Trips
7	18-24	Male	Low	Yes	1-2	Urban	High	Public Transport	Work	Time, Cost	Convenience	Regular, Short Trips
8	25-34	Female	Low	No	1-2	Urban	High	Public Transport	Work	Time, Cost	Convenience	Regular, Short Trips
9	35-44	Male	Low	Yes	2-3	Urban	High	Public Transport	Work	Time, Cost	Convenience	Regular, Short Trips
10	45-54	Female	Low	No	1-2	Urban	High	Public Transport	Work	Time, Cost	Convenience	Regular, Short Trips
11	55-64	Male	Low	Yes	2-3	Urban	High	Public Transport	Work	Time, Cost	Convenience	Regular, Short Trips
12	65+	Female	Low	No	1-2	Urban	High	Public Transport	Work	Time, Cost	Convenience	Regular, Short Trips
13	18-24	Male	Low	Yes	1-2	Urban	High	Public Transport	Work	Time, Cost	Convenience	Regular, Short Trips
14	25-34	Female	Low	No	1-2	Urban	High	Public Transport	Work	Time, Cost	Convenience	Regular, Short Trips
15	35-44	Male	Low	Yes	2-3	Urban	High	Public Transport	Work	Time, Cost	Convenience	Regular, Short Trips
16	45-54	Female	Low	No	1-2	Urban	High	Public Transport	Work	Time, Cost	Convenience	Regular, Short Trips
17	55-64	Male	Low	Yes	2-3	Urban	High	Public Transport	Work	Time, Cost	Convenience	Regular, Short Trips
18	65+	Female	Low	No	1-2	Urban	High	Public Transport	Work	Time, Cost	Convenience	Regular, Short Trips
19	18-24	Male	Low	Yes	1-2	Urban	High	Public Transport	Work	Time, Cost	Convenience	Regular, Short Trips
20	25-34	Female	Low	No	1-2	Urban	High	Public Transport	Work	Time, Cost	Convenience	Regular, Short Trips
21	35-44	Male	Low	Yes	2-3	Urban	High	Public Transport	Work	Time, Cost	Convenience	Regular, Short Trips
22	45-54	Female	Low	No	1-2	Urban	High	Public Transport	Work	Time, Cost	Convenience	Regular, Short Trips
23	55-64	Male	Low	Yes	2-3	Urban	High	Public Transport	Work	Time, Cost	Convenience	Regular, Short Trips
24	65+	Female	Low	No	1-2	Urban	High	Public Transport	Work	Time, Cost	Convenience	Regular, Short Trips
25	18-24	Male	Low	Yes	1-2	Urban	High	Public Transport	Work	Time, Cost	Convenience	Regular, Short Trips
26	25-34	Female	Low	No	1-2	Urban	High	Public Transport	Work	Time, Cost	Convenience	Regular, Short Trips
27	35-44	Male	Low	Yes	2-3	Urban	High	Public Transport	Work	Time, Cost	Convenience	Regular, Short Trips
28	45-54	Female	Low	No	1-2	Urban	High	Public Transport	Work	Time, Cost	Convenience	Regular, Short Trips
29	55-64	Male	Low	Yes	2-3	Urban	High	Public Transport	Work	Time, Cost	Convenience	Regular, Short Trips
30	65+	Female	Low	No	1-2	Urban	High	Public Transport	Work	Time, Cost	Convenience	Regular, Short Trips



Meet the Personas

Segment 1
Less Mobile, Car Reliant


Brian
I am a wheelchair user and we share the same experience. I rely on my car to get to work.

Segment 2
Young Urban Families


Betty
Driving helps me to enjoy my retirement and I rely on my car to get to work.

Segment 3
Older Less Affluent


Farah
My husband and I share our car and business community and school zone - a second car would be very handy.

Segment 4
Comfortable Empty-nesters


Gina
I work in adult social care and we just it essential for getting out and about to my clients. I don't see any need to get a car.

Segment 5
Suburban Families


Jeff
We got rid of one of our cars when we retired, with the slower pace of life, we enjoy using bus and train services.

Segment 6
Elderly And Low Income


Nigel
I work from home a lot more but still drive to work. I'm hoping that we might be able to afford a second car soon.

Segment 7
Heavy Car Users, Frequent Flyers


Oliver
I use my car every day - it's down to where I live, my job and driving the kids around.

Segment 8
Elderly And Low Income Without Cars


Pater
I miss the freedom of using my car but having an over-the-pace helps me to get out and about.

Segment 9
Urban Professionals Without Cars


Pippa
My mental condition means that bus journeys are just too uncomfortable, and I am more reliant on life from my daughter.

Segment 10
Young Low Income Without Cars


Rosa
I can get everywhere I need to easily by public transport, and I'd like to use my own car.

Segment 11
Young Low Income Without Cars


Zoe
I can access bus and Metro services from where I live but I'd love to have my own car when I can afford it.

Segment 12
Young Low Income Without Cars


Zahir
I'm out at work and struggling financially - so I walk and cycle whenever I can to save money for essentials.

Surveys on a topic

Behaviours, attitudes and circumstances

Segments

Aggregated representations of heterogeneous behaviours

UK Mobility



Meet the Personas



Segment 1 Less Mobile, Car Reliant  Brian I am a wheelchair user and the door-to-door experience is infinitely easier if I drive.	 Betty Driving helps me to enjoy my retirement and I've become reliant on it due to my arthritis.	Segment 2 Young Urban Families  Farah My husband and I share our car, and between commuting and school runs – a second car would be very handy!	Segment 3 Older Less Affluent  Gina I work in adult social care and my car is essential for getting out and about to my clients. I don't have any choice but to drive.	Segment 4 Comfortable Empty-nesters  Jeff We got rid of one of our cars when we retired. With the slower pace of life, we enjoy using bus and train services.	Segment 5 Suburban Families  Nigel I work from home a lot more but I still drive to work. I'm hoping that we might be able to afford an electric car soon.	
Segment 6 Heavy Car Users, Frequent Flyers  Oliver I use the car every day – it's down to where I live, my job and driving the kids around.	Segment 7 Elderly And Low Income Without Cars  Peter I miss the freedom of using my car but having an over-60s pass helps me to get out and about.		 Pippa My spinal condition means that bus journeys are just too uncomfortable, so I am more reliant on lifts from my daughter.	Segment 8 Urban Professionals Without Cars  Rosa I can get everywhere I need to easily by public transport, but I'd like to use my bike more.	Segment 9 Young Low Income Without Cars  Zoe I can access bus and Metro services from where I live but I'd love to have my own car when I can afford it.	 Zahir I'm out of work and struggling financially – so I walk and cycle whenever I can to save money for essentials.

Non-car Owning Segments

Non-car owners segments (7-9).
Images used in this document are of fictional people and have been chosen to portray the key characteristics of Personas.

v3 June 2023

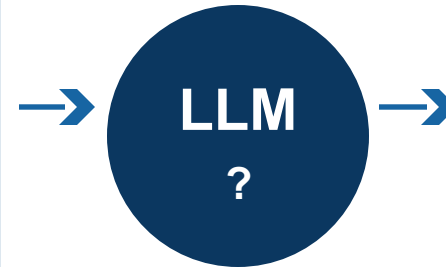
Assigning Persona to Synthetic Populations

MATSim synthetic agents

Agent attributes

Age, gender, household status and other socio-demographical information

Typically missing
Attitudes and perceptions



Pre-trained Knowledge

DfT persona

Nine travel behaviours segments

General transport habits, enablers and barriers to different transport modes & opportunities for change

Research question

Can we use LLMs to label mobility personas across different synthetic populations so that they can be used as an aggregated behavioural layer in MATSim?

y



01 Original DfT persona allocation

Clustering & discriminant analysis

Each individual has an assigned persona

[illegible]

02 Persona labelling using different LLMs

20 synthetic population attributes

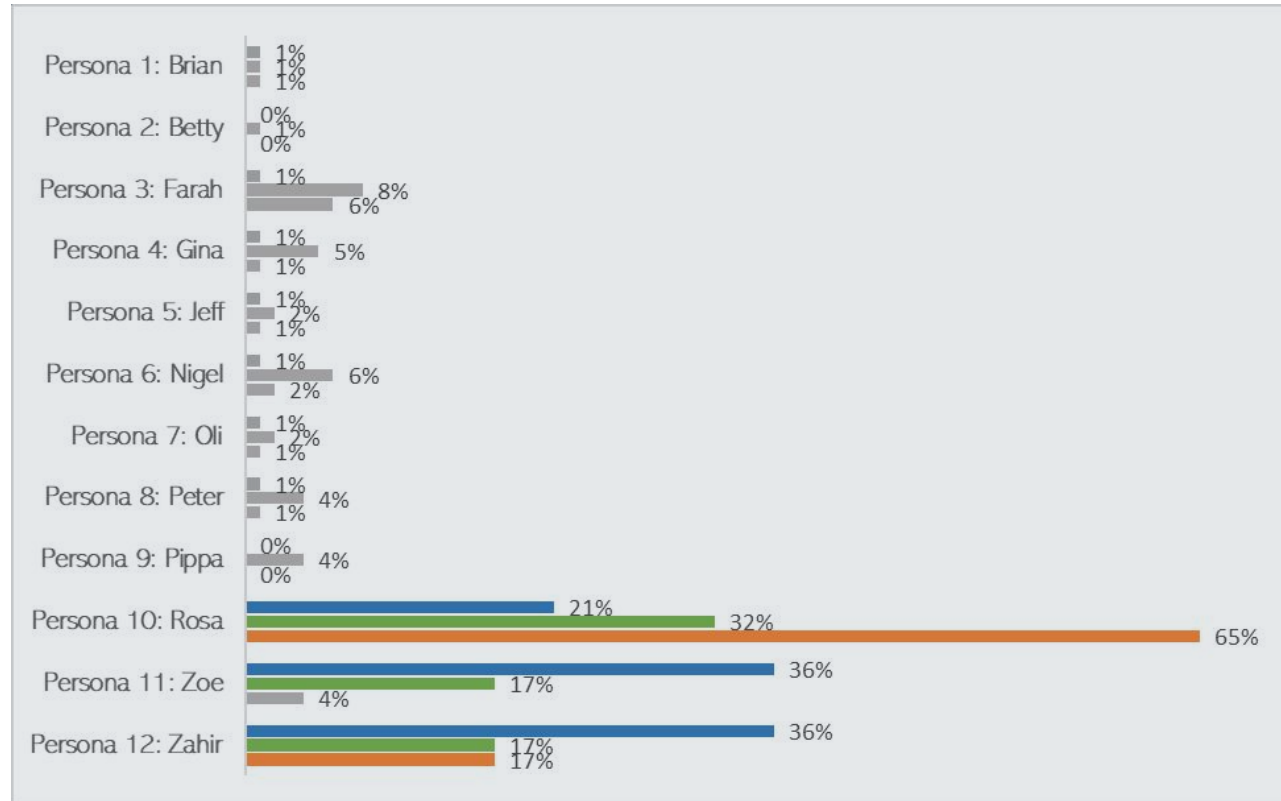
Descriptions of the mobility habits of 9 segments

Independent LLMs

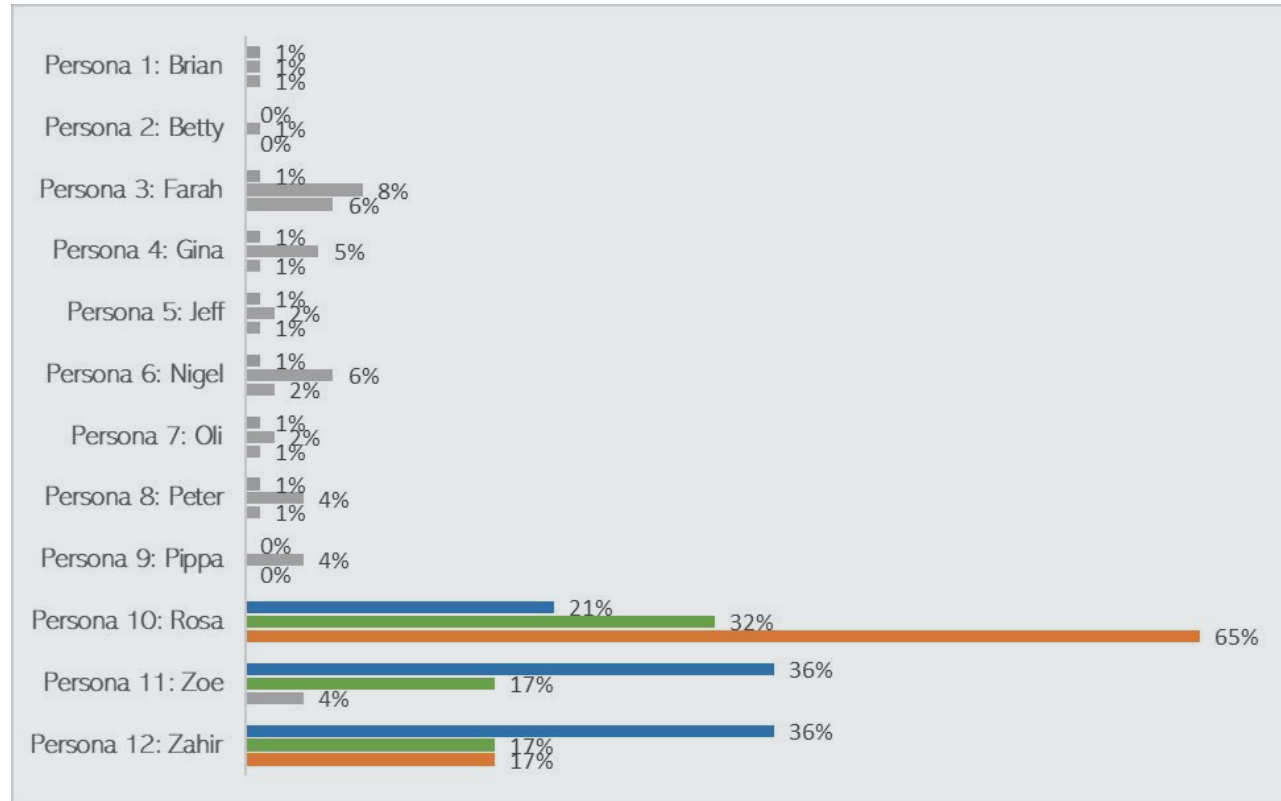


03 A probability distribution across 9 segments and a confidence level

LLM for Persona – decision fusion



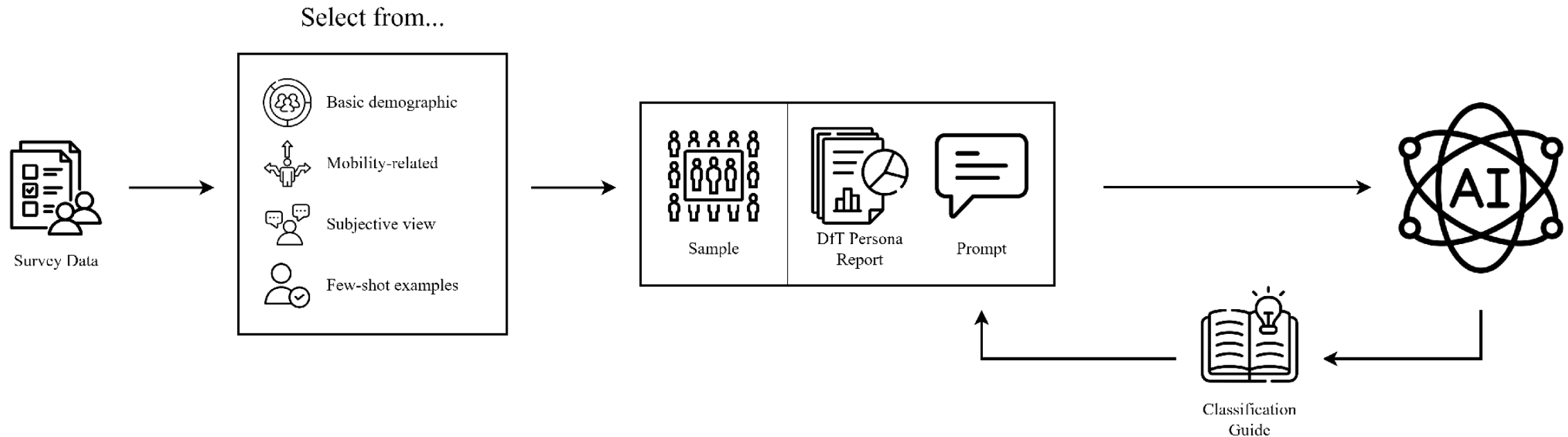
LLM for Persona – decision fusion



Confidence level

- Chat GPT 5.5 0.64
- Claude Sonnet 4.5 0.78
- Gemini 2.5 pro 0.85

LLM Prompt improvement



Add prior knowledge

Distinguish inherently similar segments for more accurate classification

LLM Prompt improvement

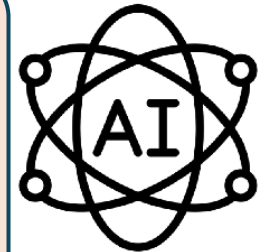
Select from...



Survey Data

Boundary rules:

- Health-related mobility constraints outweigh demographic characteristics travel primarily driven by leisure
- When mobility limitations clearly influence travel behaviour, favour Segment 1 over other car-user segments unrestricted multimodal travel
- The existence of a health condition alone is insufficient; travel behaviour must be affected



Guide



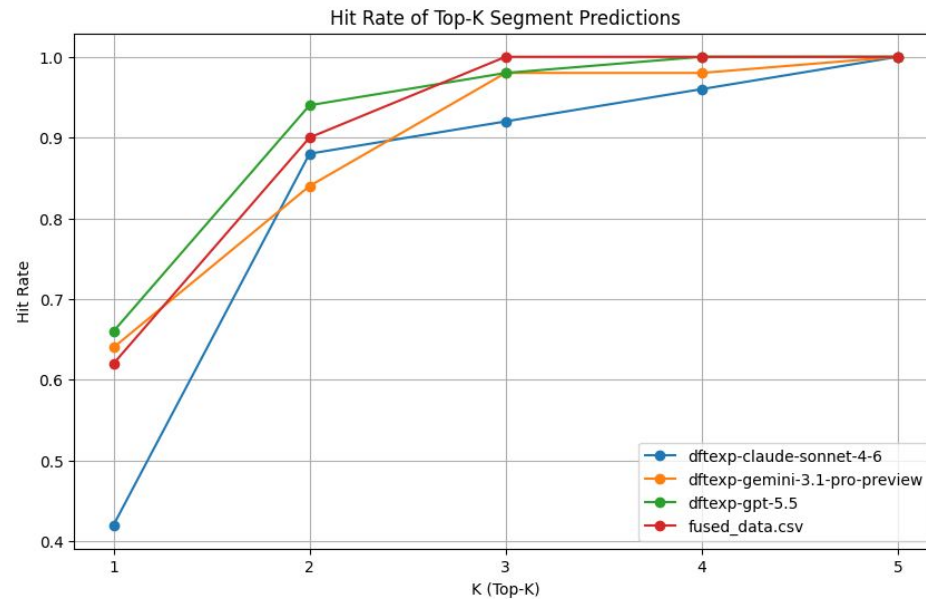
Add prior knowledge

Distinguish inherently similar segments for more accurate classification

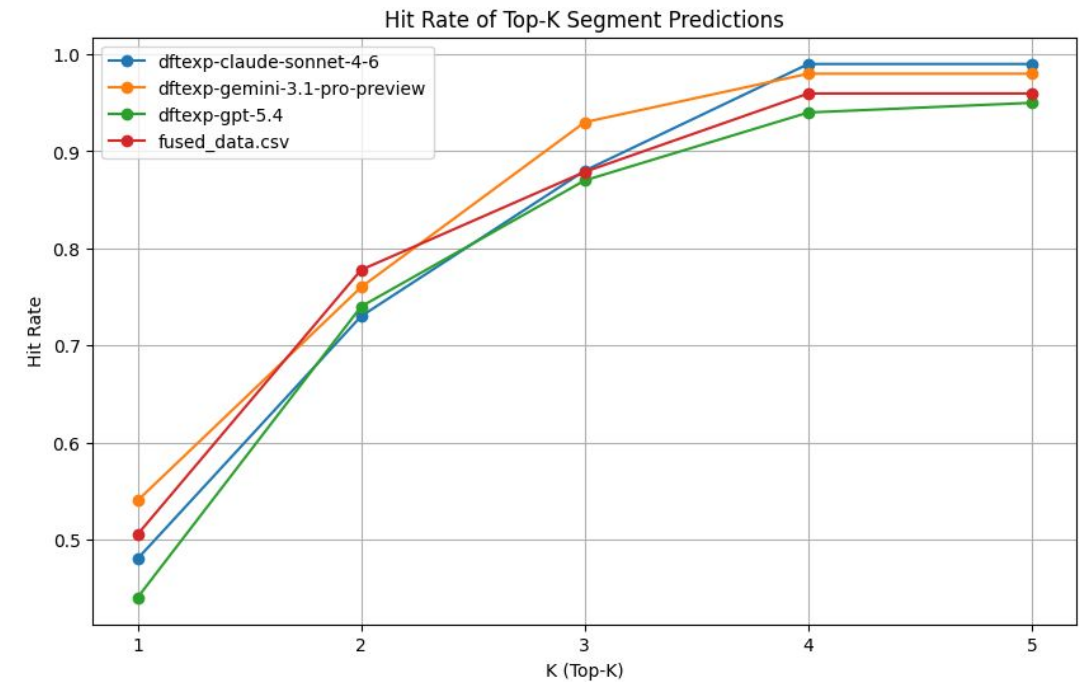
Outcome



e
Before



After



2022 accuracy

Segments 1–6: 78.9%

Segments 7–9: 91.7%

Synthetic population attributes

input



Current survey input set: 23 attributes



6

General demographic

Age, education, household cars, perceived income



5

Health and mobility limits

Difficulty walking, cycling, using buses or entering a car



3

Travel and purchasing habits

Annual driving mileage, recent flights, home delivery



9

Preferences and perceptions

Mode safety, car priorities, willingness to cycle

Future research

Mobility personas



Car-reliant traveller



Low-income traveller without a car



Urban professional without a car

Introducing interventions



MATSim
Multi-Agent Transport Simulation

LLM replanning in MATSim

Possible replanning choices



Keep driving?



Use the improved bus?



Combine walking + bus?

Thank You!



01

LLMs can understand a persona report and reproduce predictions, even using different attribute sets.

02

Accuracy is mainly affected by the fact that some segments are inherently similar.

03

With personas, (potentially) more heterogeneous MATSim agent logic reflects different individual preferences?



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